

SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE STRUCTURE

B.Tech. (Computer Science and Engineering) and Specializations: Data Science, Cyber Security
(Academic Batch 2025-2029)

Integrated B.Tech.- M.Tech. (Computer Science and Engineering)

(Academic Batch 2025-2030)



Gautam Buddha University
Gautam Budh Nagar, Greater Noida, UP, INDIA

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SEMESTER III

S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping
1	CSE201	Algorithms, Computing with Python	3	0	0	3	PCC
2	CSE202	Data Structures	3	0	0	3	PCC
3	CSE206	Database Management System	3	0	0	3	PCC
4	CSE207	Computer Organization and Architecture	3	0	0	3	PCC
5	CSE208	Operating System	3	0	0	3	PCC
6	CSE209	Security, Computing with Python Lab	0	0	4	2	PCC
7	CSE203	Web Structures Lab	0	0	4	2	PCC
8	CSE204	Database Management System Lab	0	0	4	2	PCC
9	XXXX	Set for HSMC / AIC	3	0	0	3	HSMC
Total Hours and Credits			18	0	12	24	

SEMESTER IV

S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping
1	CSE202	Design and Analysis of Algorithms	3	0	0	3	PCC
2	CSE204	Computer Networks	3	0	0	3	PCC
3	CSE206	Artificial Intelligence	3	0	0	3	PCC
4	CSE208	Theory of Automata	3	0	0	3	PCC
5	XX209	Specialization Core Course - I	3	0	0	3	PCC
6	XX209	Specialization Core Course - II	3	0	0	3	PCC
7	CSE282	Design and Analysis of Algorithms Lab	0	0	4	2	PCC
8	CSE284	Computer Networks Lab	0	0	4	2	PCC
9	XX209	Specialization Core Course - I Lab	0	0	4	2	PCC
10	XXXX	Set for HSMC / VAC	2	0	0	2	HSMC
Total Hours and Credits			20	0	12	26	

SEMESTER V

S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping
1	CSE301	Compiler Design	3	0	0	3	PCC
2	CSE302	Image Engineering	3	0	0	3	PCC
3	CSE305	High Performance Computing (HPC)	3	0	0	3	PCC
4	XX308	Specialization Core Course - III	3	0	0	3	PCC
5	XX308	Specialization Elective 1	3	0	0	3	PCC
6	XX308	Specialization Elective 2	3	0	0	3	PCC
7	XX309	Specialization Core Course - III Lab	0	0	4	2	PCC
8	XX309	Specialization Elective 1 Lab	0	0	4	2	PCC
Total Hours and Credits			18	0	8	22	

SEMESTER VI

S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping
1	CSE301	Internet of Things	3	0	0	3	PCC
2	CSE302	Cloud Infrastructure and Virtualization	3	0	0	3	PCC
3	CSE305	Engineering Aerial Vehicle System Engineering	3	0	0	3	PCC
4	XX309	Specialization Elective 3	3	0	0	3	PCC
5	XX309	Specialization Elective 4	3	0	0	3	PCC
6	XXXX	Specialization Lab	0	0	4	2	PCC
7	XX309	Specialization Core Course and Virtualization Lab	0	0	4	2	PCC
8	XX309	Specialization Elective 3 Lab	0	0	4	2	PCC
Total Hours and Credits			19	0	12	24	

SEMESTER VII

S.No.	Course Code	Course Name	L	T	P	Credits	UCC Mapping
1	CSE301	Machine Learning	3	0	0	3	PCC
2	CSE301	Machine Learning	3	0	0	3	PCC

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3	XXXX	Open Elective 3	3	0	0	3	OEC	Open Elective Course	SKB Enhancement Course (SEC)
4	XXXX	Slot for HSMC / AEC	3	0	0	3	HSMC	Humanities Social Science and Management Course	Ability Enhancement Course (AEC)
5	CSE493	Capstone Project 2*	0	0	8	4	Project	Project	Research Project (Major)
Total Hours and Credits			17	0	8	16			

SEMESTER VIII							UGC Mapping	
S.No.	Course Code	Course Name	L	T	P	Credits	AICTE MAPPING	UGC Mapping
1	CSE490	Internship II	0	0	20	10	Internship	Internship
Total Hours and Credits			0	0	20	10		Internship / Apprenticeship

* Internship will be completed by each candidate individually during the 8th semester in Industry / GRU.
The duration of the Internship will be a minimum of 4 months. It will be evaluated as per the University Examination rules.

Total Credits

168

SPECIALIZATION CORE COURSES FOR B.Tech (Computer Science and Engineering)							UGC Mapping (Reference)	
S.No.	Course Code	Course Name	L	T	P	Credits	Types	UGC Mapping (Reference)
1	CSE210	Machine Learning	3	0	0	3	Specialization Core Course-I	Major (DSG)
2	CSE286	Machine Learning Lab	0	0	4	2	Specialization Core Course-I Lab	Major (DSG)
3	CSE212	Computer Graphics and Multimedia	3	0	0	3	Specialization Core Course-II	Major (DSG)
4	CSE207	Web Technologies	3	0	0	3	Specialization Core Course-III	Major (DSG)
5	CSE381	Web Technologies Lab	0	0	4	2	Specialization Core Course-III Lab	Major (DSG)

SPECIALIZATION CORE COURSES FOR B.Tech CSE Specialization: Data Science							UGC Mapping (Reference)	
S.No.	Course Code	Course Name	L	T	P	Credits	Types	UGC Mapping (Reference)
1	DSI210	Machine Learning	3	0	0	3	Specialization Core Course-I	Major (DSG)
2	DSI286	Machine Learning Lab	0	0	4	2	Specialization Core Course-I Lab	Major (DSG)
3	DSI217	Data Mining and Warehousing	3	0	0	3	Specialization Core Course-II	Major (DSG)
4	DSI307	Big Data Analytics	3	0	0	3	Specialization Core Course-III	Major (DSG)
5	DSI351	Big Data Analytics Lab	0	0	4	2	Specialization Core Course-III Lab	Major (DSG)

SPECIALIZATION CORE COURSES FOR B.Tech CSE Specialization: Cyber Security							UGC Mapping (Reference)	
S.No.	Course Code	Course Name	L	T	P	Credits	Types	UGC Mapping (Reference)
1	CSY210	Cryptography and Network Security	3	0	0	3	Specialization Core Course-I	Major (DSG)
2	CSY286	Cryptography and Network Security Lab	0	0	4	2	Specialization Core Course-I Lab	Major (DSG)
3	CSY212	Introduction to Cyber Security	3	0	0	3	Specialization Core Course-II	Major (DSG)
4	CSY307	Ethical Hacking and Penetration Testing	3	0	0	3	Specialization Core Course-III	Major (DSG)
5	CSY381	Ethical Hacking and Penetration Testing Lab	0	0	4	2	Specialization Core Course-III Lab	Major (DSG)

PROGRAM ELECTIVES FOR B.Tech CSE								
S.No.	Course Code	Course Name	L	T	P	Credits	Types	
1	CSE209	Soft Computing	3	0	0	3	Program Elective-1	
2	CSE311	Introduction to Cyber Security	3	0	0	3		
3	CSE313	Game Design and Development	3	0	0	3		
4	CSE315	Information Retrieval Systems	3	0	0	3		
5	CSE383	Soft Computing Lab	0	0	4	2	Program Elective Lab-1	
6	CSE385	Introduction to Cyber Security Lab	0	0	4	2		
7	CSE387	Game Design and Development Lab	0	0	4	2		
8	CSE389	Information Retrieval Systems Lab	0	0	4	2		
9	CSE317	Optimization Techniques	3	0	0	3	Program Elective-2	
10	CSE319	Data Mining and Warehousing	3	0	0	3		
11	CSE321	Evolutionary Computing	3	0	0	3		
12	CSE323	Parallel and Distributed Computing	3	0	0	3		
13	CSE325	Computer Vision	3	0	0	3	Program Elective-3	
14	CSE316	Artificial Language Processing	3	0	0	3		
15	CSE318	Wireless Communication Networks	3	0	0	3		
16	CSE314	Augmented and Virtual Reality	3	0	0	3		
17	CSE320	Blockchain Technology	3	0	0	3	Program Elective-4	
18	CSE318	Automation and Robotics	3	0	0	3		
19	CSE320	Cloud Computing	3	0	0	3		
20	CSE322	Digital Twin Systems	3	0	0	3		

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PROGRAM ELECTIVES FOR B.TECH CSE SPECIALIZATION - DATA SCIENCE							Credits			Types
S.No.	Course Code	Course Name	L	T	P					
1	DS1205	Data Engineering and Visualization	3	0	0					Program Elective-1
2	DS1311	Time Series Analysis and Forecasting	3	0	0					
3	DS1313	Multivariate Data Analysis	3	0	0					
4	DS1315	Information Retrieval and Visualization	3	0	0					Program Elective-2
5	DS1316	Data Engineering and Visualization Lab	0	0	4					
6	DS1318	Time Series Analysis and Forecasting Lab	0	0	4					
7	DS1387	Multivariate Data Analysis Lab	0	0	4					Program Elective-3
8	DS1389	Information Retrieval Systems Lab	0	0	4					
9	DS1390	Optimization Methods in Data Science	3	0	0					
10	DS1319	Sequential Modeling and Analytics	3	0	0					Program Elective-4
11	DS1321	Causal Discovery and Explainable Analytics	3	0	0					
12	DS1322	Recommendation Systems	3	0	0					
13	DS1308	Web Analytics	3	0	0					Program Elective-3
14	DS1310	Business Intelligence and Analytics	3	0	0					
15	DS1312	Geospatial Analytics	3	0	0					
16	DS1314	Graph Analytics	3	0	0					Program Elective-4
17	DS1316	Healthcare Analytics	3	0	0					
18	DS1318	Computational Neuroscience	3	0	0					
19	DS1320	Data Privacy and Security Analytics	3	0	0					Program Elective-4
20	DS1322	Responsible Data Science and Ethics	3	0	0					

PROGRAM ELECTIVES FOR B.Tech. CSE Specialization: Cyber Security						
S.No.	Course Code	Course Name	L	T	P	Credits
1	CSY205	Secure Programming and Web Security	3	0	0	3
2	CSY212	Operating System Security	3	0	0	3
3	CSY213	Intrusion Detection and Prevention Systems	3	0	0	3
4	CSY215	Blockchain Security	3	0	0	3
5	CSY232	Secure Programming and Web Security Lab	0	0	4	2
6	CSY235	Operating System Security Lab	0	0	4	2
7	CSY237	Intrusion Detection and Prevention Systems Lab	0	0	4	2
8	CSY239	Blockchain Security Lab	0	0	4	2
9	CSY247	Data Ware Analysis and Reverse Engineering	3	0	0	3
10	CSY249	Cloud Architecture and Security	3	0	0	3
11	CSY251	Privacy Preserving Technologies	3	0	0	3
12	CSY253	Security Information and Event Management	3	0	0	3
13	CSY255	Access Control and Identity Management	3	0	0	3
14	CSY257	Secure Software Engineering	3	0	0	3
15	CSY262	Digital Forensics	3	0	0	3
16	CSY264	AI for Cyber Security	3	0	0	3
17	CSY266	Enterprise Security and Management	3	0	0	3
18	CSY268	Mobile and Wireless Sensor Network Security	3	0	0	3
19	CSY270	Security Intelligence and Cyber Analytics	3	0	0	3
20	CSY272	Cloud Native and DevSecOps	3	0	0	3

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